

Evolution of the human diet: linking our ancestral diet to modern functional foods as a means of chronic disease prevention.

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The evolution of the human diet over the past 10,000 years from a Paleolithic diet to our current modern pattern of intake has resulted in profound changes in feeding behavior. Shifts have occurred from diets high in fruits, vegetables, lean meats, and seafood to processed foods high in sodium and hydrogenated fats and low in fiber. These dietary changes have adversely affected dietary parameters known to be related to health, resulting in an increase in obesity and chronic disease, including cardiovascular disease (CVD), diabetes, and cancer. Some intervention trials using Paleolithic dietary patterns have shown promising results with favorable changes in CVD and diabetes risk factors. However, such benefits may be offset by disadvantages of the Paleolithic diet, which is low in vitamin D and calcium and high in fish potentially containing environmental toxins. More advantageous would be promotion of foods and food ingredients from our ancestral era that have been shown to possess health benefits in the form of functional foods. Many studies have investigated the health benefits of various functional food ingredients, including omega-3 fatty acids, polyphenols, fiber, and plant sterols. These bioactive compounds may help to prevent and reduce incidence of chronic diseases, which in turn could lead to health cost savings ranging from \$2 to \$3 billion per year as estimated by case studies using omega-3 and plant sterols as examples. Thus, public health benefits should result from promotion of the positive components of Paleolithic diets as functional foods.